Roles & Responsibilities

Requirements Engineering Team

Governing the Holonic Interface — Who Owns What, and Why

Purpose of This Document

This document defines the role boundaries, responsibilities, scope of authority, and collaboration protocols for the Requirements Engineering (RE) Gateway Team and its counterpart teams within the Knowledge Intelligence Holarchy (KIH). It applies the INCOSE systems architecture principle of separating interface authority from component authority to establish a durable, unambiguous governance model for the holonic requirements engineering function.

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Next Steps: (1) Domain Governance Team review of ORSD co-ownership boundaries | (2) CoE review of execution responsibilities | (3) Practice Leader input on persona taxonomy completeness | (4) Integration with RDSG v2.0 STPA/STAMP Edition — persona-holon lifecycle alignment

1. Governing Principle: Constitutional Authority

The Requirements Engineering Gateway Team holds constitutional authority over the holonic interface — not content authority over the holons themselves. This distinction is foundational to every role boundary defined in this document.

The Constitutional Principle

The RE Gateway Team writes the constitution that all holons must conform to. They do not author the holons themselves.

A Supply Chain Domain Holon is defined by the Supply Chain Domain Governance Team — but it cannot participate in the holarchy until the RE Gateway Team certifies that it correctly implements the PEP contract and binds cleanly to the RCB assembly protocol.

This mirrors the INCOSE Systems Architecture principle of separating interface authority from component authority. The Gateway team owns the interface; the domain teams own the components.

In INCOSE terms, the RE Gateway Team is the Systems Architect of the L2-to-L3 interface. They own the architecture, the interface contracts, the requirements traceability model, and the trust metrology. They do not own the content that flows through what they govern.

2. Teams Covered by This Document

Four organizational actors share responsibility for requirements engineering within the KIH. Their boundaries are distinct, their collaboration points are formal, and their authority domains do not overlap.

Team	Primary Function	INCOSE Analogy
RE Gateway Team	Owns the holonic interface specification — PEP contracts, RCB binding protocol, AcceptanceCriteria for Simplex Gate, persona taxonomy, requirements traceability model, and ISO standard-to-constraint mappings	Systems Architect — L2-to-L3 Interface Control Document owner
Domain Governance Teams	Own the domain-specific holon content — vocabulary, SHACL invariant specifications, SWRL rule declarations, data contract quality SLAs, and domain ontology definitions	Component Authority — accountable for individual system elements
Centre of Excellence (CoE)	Executes the inbound admission pipeline; maintains the ORSD library; monitors MTBH metrics; implements RBAC permission bindings; validates ISO conformance claims at Steps 1–4	Integration & Verification — system integration and test function
Practice Leaders	Provide practitioner-grounded stakeholder needs; input to persona taxonomy; co-own ORSD domain content; define what knowledge practitioners actually need governed	Stakeholder Representative — upstream needs authority

3. RE Gateway Team: Detailed Responsibilities

The following sections define the five core responsibility domains of the RE Gateway Team in full. Each section specifies what the team owns, what it explicitly does not own, and how its authority connects to INCOSE requirements management best practice.

3.1 Holonic Interface Specification (PEP Contracts & RCB Protocol)

What the RE Gateway Team Owns

The RE Gateway Team owns the holonic interface specification in its entirety. This means they define and enforce the contract that any holon must satisfy to be composable with the gateway. Concretely, this specification includes:

- The PEP contract schema — what each Policy Enforcement Point (Compliance, Risk, Legal, Security, Data Governance, and AI Governance PEPs) must declare for a holon to be admissible to the holarchy
- The RCB dimension binding protocol — how a holon registers its WHO/WHAT/WHEN/WHERE/WHY context surface so that Stage 0 Requirement Resolution can build a valid Requirement Context Bundle from it
- The AcceptanceCriteria specification for the Simplex Gate — the testable conditions a holon's outputs must satisfy before the Simplex Action Gateway will permit any Irreversible Zone 6 action it triggers
- The interface versioning policy — how changes to the PEP contract or RCB protocol are versioned, communicated, and migrated across conforming domain holons

What the RE Gateway Team Does Not Own

- The content of any holon — that is the Domain Governance Team's authority
- The admission pipeline execution — that is the CoE's execution responsibility
- The internal ontology structure of domain vocabularies — domain teams own their SHACL invariants and SWRL rule declarations

> **INCOSE Connection: Interface Control Document (ICD) Ownership**
>
> The RE Gateway Team holds the ICD for the L2-to-L3 interface. In INCOSE practice, the ICD owner has formal authority over the interface specification and sign-off rights on any component that crosses that interface. A domain holon that does not satisfy the PEP contract is, in INCOSE terms, a non-conforming component — it may not be integrated until the interface violation is resolved.

3.2 ORSD Template Structure (Gateway-Visible Sections)

Co-Ownership with Domain Governance Teams

The Operational Requirements Specification Document (ORSD) template is jointly governed. The RE Gateway Team and Domain Governance Teams co-own different sections of the template, with a formal sign-off protocol that prevents either party from unilaterally defining the full specification.

ORSD Section	Content	Owner / Sign-off Authority
PEP scope declarations	Which PEPs are activated by this holon, what compliance scope they cover, and what the PEP verdict triggers	RE Gateway Team — formal sign-off required
5D context surface	WHO personas served, WHAT concepts governed, WHEN temporal rules carried, WHERE domain boundary occupied, WHY intent-tags responded to	RE Gateway Team — formal sign-off required
AcceptanceCriteria stubs (Simplex Gate)	The testable condition stubs required for Simplex Gate validation — populated by domain teams, validated by Gateway team	RE Gateway Team — formal sign-off required
Domain vocabulary	The actual ontology terms, synonyms, and concept definitions used by this holon	Domain Governance Team owns; CoE maintains library
SHACL invariant specifications	The specific SHACL shape constraints that govern this domain's data quality and structural requirements	Domain Governance Team owns
SWRL rule declarations	The behavioral rules that fire on this domain's requirement class membership	Domain Governance Team owns
Data contract quality SLAs	Performance, reliability, and data quality thresholds for this domain's knowledge assets	Domain Governance Team owns

In practice, the RE Gateway Team participates in ORSD template reviews as the gateway-interface authority. They have a formal sign-off role on the PEP contract sections and the 5D context surface declarations — but not on the ontology content sections.

3.3 Persona Models and WHO-Dimension Governance

Strongest Direct Ownership

The persona model is the WHO dimension of the holarchy — it governs access scope, PEP permission sets, response format, and MTBH_WHO fidelity measurement. The RE Gateway Team must own this because persona misrouting is a gateway-level failure, not a domain-level failure.

RE Gateway Team Owns

- **Persona taxonomy:** Field Consultant, Senior Consultant, Engagement Manager, Account Manager, Sales, Practice Lead, Executive, Machine Agent — each as a governed persona-holon with formal identity in the Identity Ontology
- **RBAC permission bindings per persona:** which IP assets, which domain holons, and which PEP compliance scopes each persona may access

- **MTBH_WHO monitoring thresholds and trip conditions:** the dimensional fidelity targets for the WHO circuit breaker dimension, and the conditions that trigger a circuit breaker trip at the persona level
- **Persona-holon lifecycle protocol:** how new roles are onboarded, how persona-holons are versioned when org structure changes, and how deprecated persona-holons are retired from active gateway queries

RE Gateway Team Does Not Own

- The content of role-specific training, which belongs to role interface teams in collaboration with practice leaders
- The experiential design of persona-facing interfaces (e.g., Jarvis interaction design) — that belongs to the role of interface teams

INCOSE Connection: Stakeholder Needs Register

INCOSE treats stakeholder needs as the upstream driver of requirements decomposition. Persona models in the KIH are the architectural equivalent of the stakeholder needs register — they are the formal specification of who the system must serve, and every downstream requirement (PEP scope, SHACL constraint, SWRL rule) must be traceable to a persona-holon.

The RE Gateway Team is therefore the requirements traceability anchor for the WHO dimension. Every SHACL shape enforced by the gateway must be traceable to a persona-holon WHY intent, or it has no legitimate basis in the requirements hierarchy.

3.4 Requirements Engineering Pipeline (INCOSE RM + ISO)

Most Consequential and Least Obvious Responsibility

The inbound admission pipeline is not a data engineering pipeline — it is a requirements engineering pipeline. This is the area where the Gateway team's role is most consequential and least obvious. Every knowledge asset admitted to the holarchy must satisfy requirements that were derived from stakeholder needs (persona models), expressed as SHACL invariants, validated against STPA/STAMP UCA analysis, and governed by ISO standards. The RE Gateway Team must own the requirements engineering layer of that pipeline.

INCOSE RM Four-Level Traceability

The RE Gateway Team owns the requirements traceability model that links the following four levels. This model is the INCOSE RM best practice applied to a knowledge holarchy rather than a physical system:

Level	INCOSE RM Layer	KIH Equivalent	Artifact / Record
L1	**Stakeholder Need**	Persona-holon WHY intent — what the practitioner needs the knowledge to accomplish	Persona-holon individual in Identity Ontology
L2	**System Requirement**	SHACL constraint on the knowledge holon — the engineering specification of what must be true of any admitted asset	ORSD — System requirements document

L3	**Verification Criterion**	SHACL validation pass/fail in the admission pipeline — the testable acceptance criterion for each requirement	KGCL change record — change control log
L4	**Validation Measure**	MTBH dimensional score in production — operational verification that the admitted asset is serving the stakeholder need it was admitted to serve	Circuit Breaker trip record — nonconformance report

ISO Standard Mappings

The RE Gateway Team owns the mapping of ISO quality requirements to gateway-level enforcement mechanisms. Domain teams declare ISO conformance claims in their ORSDs; the Gateway team validates that those claims are testable and enforces them at Steps 1–4.

ISO Standard	Domain	Gateway Enforcement Mechanism
ISO 8000	Data quality	Mapped to SHACL constraints on knowledge holon data quality dimensions; enforced at Steps 1–4 of the admission pipeline
ISO/IEC/IEEE 24765	Systems and software engineering vocabulary	Governs the core vocabulary holons that all domain terminology must be conformant with; the Gateway team validates vocabulary conformance during holon certification
ISO/IEC 25010	System quality model	Maps quality characteristics (functionality, reliability, usability, efficiency, maintainability, portability) to MTBH dimensional scores as operational verification criteria; Gateway team owns this mapping

Practitioner Knowledge Requirements Elicitation

The RE Gateway Team defines the requirements specification that every inbound practitioner's knowledge asset must satisfy before it can become a Knowledge Holon. This is not the Governance Agent Audit — that is CoE execution. This is the upstream requirements definition work that specifies what the audit is checking against.

Example: Supply Chain Framework Requirements Elicitation

Practitioner stakeholder need: 'A supply chain framework must carry the client industry, engagement outcome, and geographic scope before it can be used to generate client-facing proposals.'

RE Gateway Team translation: This stakeholder need is formalized as a SHACL shape (sh:SupplyChainFrameworkShape) with three required properties (client_industry, engagement_outcome, geographic_scope) and a PEP compliance rule specifying that any engagement output derived from this framework must cite these three properties in its compliance attestation.

The INCOSE RM discipline is precisely what prevents the holarchy from becoming a governed but useless archive — requirements traceability from practitioner need to SHACL constraint to MTBH metric is what ensures the knowledge being ingested is the knowledge practitioners actually need, governed in the way their work actually demands.

3.5 Trust Metrology and MTBH Governance

The RE Gateway Team owns the trust metrology architecture — the MTBH (Mean Time Between Holonic failures) dimensional framework that provides operational verification that the governance structure is performing as designed. This is not monitoring work; it is the architectural definition of what constitutes a governance failure at each dimension.

MTBH Dimension	Circuit Breaker Axis	What RE Gateway Team Owns	What CoE Executes
MTBH_WHO	WHO dimension	Monitoring thresholds, trip conditions, persona misrouting failure definition	Real-time monitoring; trip event logging; escalation routing
MTBH_WHAT	WHAT dimension	Ontology coverage thresholds; unresolved entity failure definition; acceptable disambiguation rate	OntologyGap detection; disambiguation rate tracking
MTBH_WHEN	WHEN dimension	Temporal constraint violation definition; TMP rule timeout calibration thresholds; SLA breach definition	TMP rule monitoring; timeout event logging
MTBH_WHERE	WHERE dimension	Domain boundary violation definition; cross-domain access failure definition; Zone 6 audit scope	Domain boundary enforcement; audit log maintenance
MTBH_WHY	WHY dimension	Intent-tag misalignment failure definition; Human-System Misalignment hazard threshold; Requirements Review trigger conditions	JTBD mismatch detection; Requirements Review queue management

4. Responsibility Assignment Matrix

The following matrix summarizes ownership and participation across all key responsibilities. Symbol definitions appear in the legend below.

O	C	I	E	V	S	M

O Owns · C Co-owns · I Input · E Executes · V Validates · S Supports · M Monitors · — Not involved

Responsibility	RE Gateway	Domain Gov	CoE	Practice Leaders
INTERFACE & CONTRACT GOVERNANCE				
Holonic interface specification (PEP contracts)	O	—	E	—
RCB dimension binding protocol definition	O	—	E	—
Acceptance Criteria spec for Simplex Gate	O	I	E	—
Interface versioning and migration policy	O	I	E	—
Holon certification (PEP contract conformance)	O	—	E	—
ORSD TEMPLATE GOVERNANCE				
ORSD template structure — gateway-visible sections	C	C	S	—
PEP scope declarations (sign-off authority)	O	I	—	—
5D context surface declarations (sign-off authority)	O	I	—	—
Acceptance Criteria stubs (sign-off authority)	O	I	—	—
Domain vocabulary and ontology content sections	—	O	S	I
SHACL invariant specifications	—	O	V	—
SWRL rule declarations	—	O	V	—
Data contract quality SLAs	—	O	S	I
PERSONA MODEL & WHO-DIMENSION GOVERNANCE				
Persona taxonomy definition	O	—	I	I
RBAC permission bindings per persona	O	I	E	I
MTBH_WHO monitoring thresholds and trip conditions	O	—	M	—
Persona-holon lifecycle protocol	O	—	E	I
Role-specific training content	—	—	—	O
Persona-facing interface design (Jarvis UX)	—	—	—	O

REQUIREMENTS TRACEABILITY & PIPELINE GOVERNANCE				
Requirements traceability model (Need→Constraint→Metric)	O	I	S	S
ISO standard → SHACL constraint mapping	O	I	V	—
Practitioner knowledge requirements elicitation	S	I	—	O
SHACL invariant derivation from practitioner needs	O	I	S	I
Admission pipeline execution	—	—	E	—
Governance Agent Audit execution	—	—	E	—
ORSD library maintenance	—	—	E	—
KGCL change record management	—	—	E	—
TRUST METROLOGY & MTBH GOVERNANCE				
MTBH dimensional framework definition	O	—	—	—
Circuit breaker trip condition definitions	O	—	—	—
MTBH dimensional monitoring (all 5 dimensions)	—	—	M	—
Requirements Review trigger conditions	O	—	E	—
Temporal constraint calibration (TMP rule thresholds)	O	—	S	—
NonConformance report (Circuit Breaker trip record)	O	—	E	—

5. Collaboration Protocols and Decision Rights

The following table defines the formal touchpoints between the RE Gateway Team and other teams — when each party has decision rights, when they have input rights, and when a formal sign-off is required before work can proceed.

Interaction / Touchpoint	RE Gateway Team Authority	Other Team Authority	Escalation / Resolution Path
Domain holon onboarding — new holon requests admission to the holarchy	Formal certification sign-off on PEP contract conformance and RCB binding. Without Gateway sign-off, the holon is not admitted.	Domain Governance Team defines the holon content and declares conformance in the ORSD.	If PEP contract defects are found, Domain team must remediate before resubmission. CoE tracks remediation status.
ORSD template review — gateway-visible sections	Formal sign-off on PEP scope, 5D context surface, and AcceptanceCriteria stubs. No sign-off = ORSD not activated.	Domain Governance Team owns the ontology content sections. Practice Leaders provide stakeholder needs input.	Disagreements on the scope of gateway-visible vs. domain-content sections escalate to a joint Architecture Review. CoE facilitates.
New persona role onboarding	Owns the definition of the persona-holon and	Practice Leaders provide role definition	Organizational restructuring that changes

	its RBAC permission bindings. Formal activation required before gateway routes requests to the new persona.	input. CoE implements RBAC bindings in the Identity Ontology.	persona taxonomy requires Gateway team review before activation. HR input is advisory; Gateway authority is final on RBAC scope.
ISO conformance claim in ORSD	Validates that the ISO conformance claim is testable and enforces it at Steps 1–4. Can reject claims that are not expressed as testable SHACL constraints.	Domain Governance Team declares the conformance claim. CoE validates the claim's implementation against the Gateway team's enforcement specification.	Untestable conformance claims are returned to the Domain team with a defect notice. Resolution requires the claim to be expressed as a SHACL shape with a SPARQL test query.
Circuit breaker trip event (any MTBH dimension)	Owns the nonconformance report. Determines whether the trip indicates a requirements specification defect, an AI capability defect, or an operational calibration issue.	CoE executes the monitoring that detected the trip. Domain team provides domain context for root cause analysis.	Gateway team triages within defined SLA. Root cause (a) → Requirements Review; root cause (b) → development backlog; root cause (c) → threshold calibration update.
Practitioner knowledge elicitation (new framework, methodology, or IP asset for ingestion)	Facilitates the requirements derivation process: translates practitioner-stated needs into formal SHACL invariants and PEP compliance rules.	Practice Leaders own the practitioner's needs. CoE executes the admission pipeline once requirements are specified.	If the practitioner's need cannot be expressed as a testable SHACL constraint, it is logged as an OntologyGap candidate and escalated to the ontology extension process.

6. Explicit Boundary: What the RE Gateway Team Does Not Own

Clarity on what the RE Gateway Team does not own is as important as defining what it does own. Boundary violations in either direction — Gateway team encroaching on domain content, or domain teams encroaching on interface specifications — are governance failures that the Requirements Traceability Model will detect.

What RE Gateway Team Does NOT Own	Who Owns It	Why the Boundary Is Here
Holon content — the actual knowledge, vocabulary, or ontology definitions inside any domain holon	**Domain Governance Teams**	Component authority is separate from interface authority (INCOSE principle). Gateway team owning content would create governance capture — the interface authority should not also define what flows through it.
Admission pipeline execution — running the Governance Agent Audit, processing ORSD submissions	**CoE**	Execution of a verified process is separate from the specification of that process. The Gateway team specifies what the audit checks; the CoE executes the check.
Role-specific training content and experiential interface design	**Practice Leaders and Role Interface Teams**	Training and UX design are delivery activities that require domain expertise and stakeholder empathy, which the Gateway

		team does not hold. These belong to the teams closest to the practitioners.
Domain ontology extension decisions — whether to add new concepts to a domain ontology	**Domain Governance Teams, with OntologyGap input from Gateway**	Ontology content belongs to the domain. The Gateway team can raise an OntologyGap finding; it cannot unilaterally extend the ontology.
Individual holon data quality remediation — fixing a specific data quality defect in an admitted holon	**Domain Governance Teams**	Data quality remediation is a content-level operation. The Gateway team defines the quality threshold (SHACL constraint); it does not perform the remediation.
Practitioner-facing workflow design within domain applications	**Domain Governance Teams and Practice Leaders**	Application workflow design is a domain concern. The Gateway team governs what the workflow must produce at the interface boundary; it does not govern how the workflow is structured inside the domain.

7. Summary: The RE Gateway Team's Architectural Identity

The RE Gateway Team is, in INCOSE terms, the Systems Architect of the L2-to-L3 interface. They own four things that no other team owns: the architecture of the interface through which all knowledge holons must pass; the interface contracts that all holons must satisfy to be composable; the requirements traceability model that connects practitioner need to SHACL constraint to MTBH metric; and the trust metrology that provides operational verification that the governance structure is performing as designed.

They do not own the content that flows through what they govern. That is not a limitation — it is the architectural guarantee that makes the holarchy trustworthy. A governance layer that also controls what it governs is not governance; it is capture. The RE Gateway Team's authority is powerful precisely because it is limited to the interface.

The INCOSE Systems Architecture Principle in Practice

Separating interface authority from component authority is the foundational principle that makes complex system integration manageable at scale. When the interface authority and component authority are the same party, interface evolution is driven by content interests rather than system-wide composability needs.

In the KIH, the RE Gateway Team holding the ICD for the L2-to-L3 interface means that every domain holon — regardless of how mature, how complex, or how strategically important — must satisfy the same interface specification to participate in the holarchy. That uniformity of interface is what makes holonic composability possible and what makes the trust metrology meaningful.

The ORSD is the system requirements document. The KGCL change record is the change control log. The Circuit Breaker trip record is the nonconformance report. The RE Gateway Team holds all three — not because they execute them, but because they own the requirements engineering framework that gives them their authority.